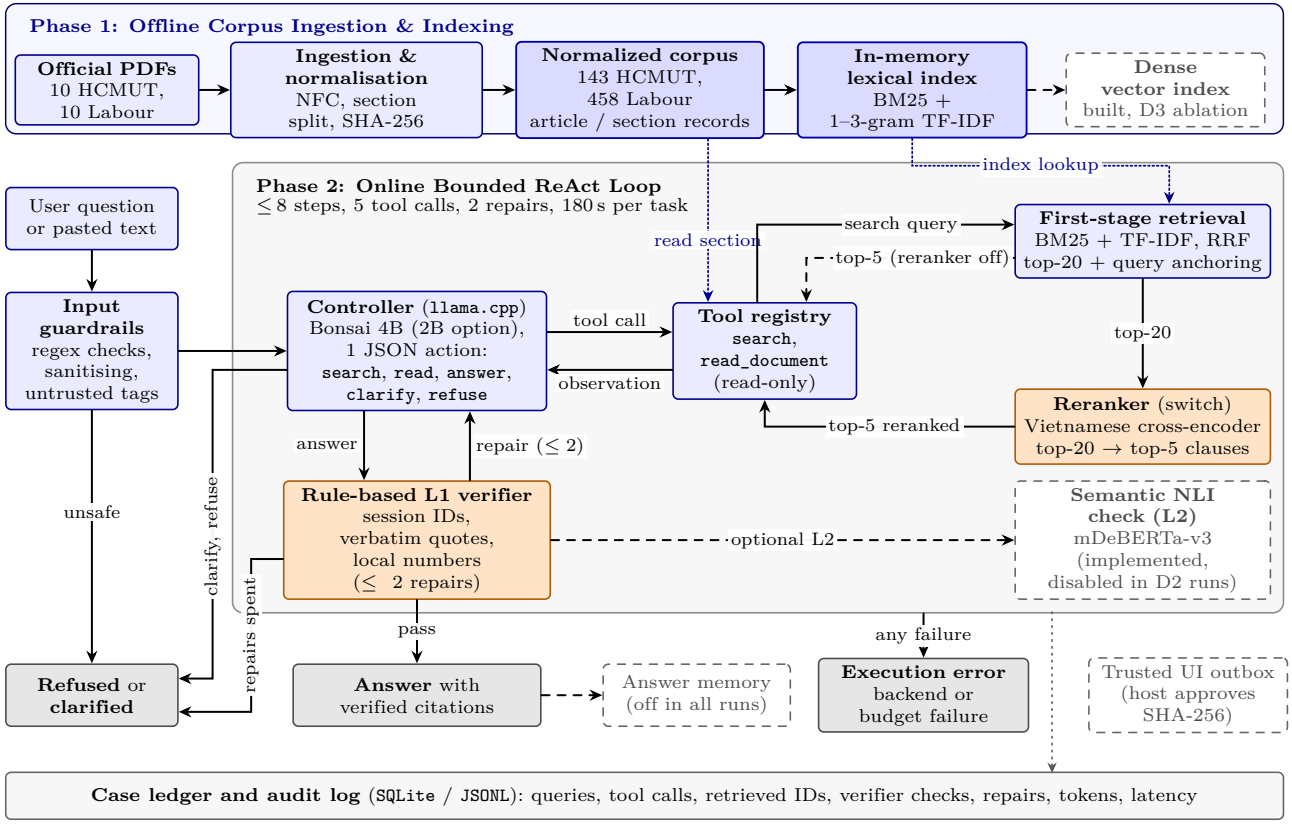

Phase 1: Offline Corpus Ingestion & Indexing
Official PDFs
10 HCMUT,
10 Labour
Ingestion & normalisation
NFC, section split, SHA-256
Normalized corpus
143 HCMUT,
458 Labour
article / section records
In-memory lexical index
BM25 +
1–3-gram TF-IDF
Dense vector index
built, D3 ablation
index lookup
Phase 2: Online Bounded ReAct Loop
≤ 8 steps, 5 tool calls, 2 repairs, 180 s per task
User question or pasted text
Input guardrails
regex checks, sanitising, untrusted tags
read section
search query
top-5 (reranker off)
First-stage retrieval
BM25 + TF-IDF, RRF
top-20 + query anchoring
top-20
Controller (llama.cpp)
Bonsai 4B (2B option),
1 JSON action:
search, read, answer,
clarify, refuse
tool call
observation
Tool registry
search,
read_document
(read-only)
top-5 reranked
Reranker (switch)
Vietnamese cross-encoder
top-20 → top-5 clauses
answer
repair (≤ 2)
unsafe
clarify, refuse
repairs spent
Rule-based L1 verifier
session IDs,
verbatim quotes,
local numbers
(≤ 2 repairs)
optional L2
Semantic NLI check (L2)
mDeBERTa-v3
(implemented,
disabled in D2 runs)
pass
any failure
Refused or clarified
Answer with verified citations
Answer memory
(off in all runs)
Execution error
backend or budget failure
Trusted UI outbox
(host approves
SHA-256)
Case ledger and audit log (SQLite / JSONL): queries, tool calls, retrieved IDs, verifier checks, repairs, tokens, latency